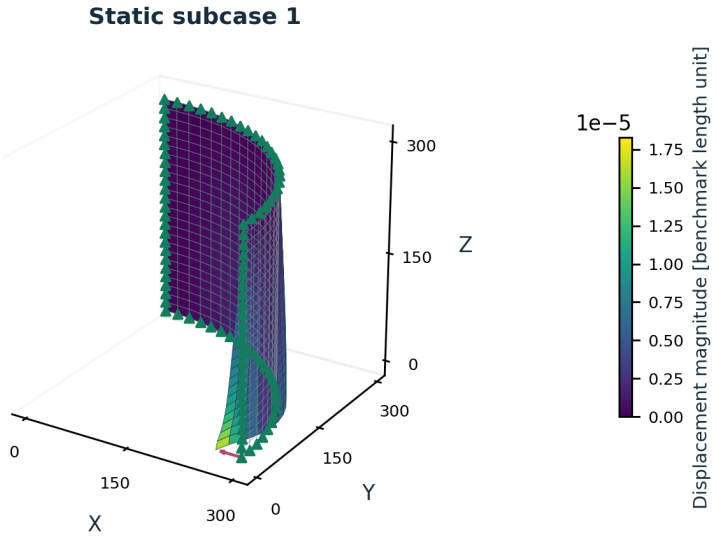
Pinched cylinder: refined octant

MH_pinched_cylinder_refined | SOL 101 | Published shell benchmark

Same cylinder geometry and load as coarse companion; 24x24 octant mesh.

Geometry / mesh	625 nodes; 576 elements. Same cylinder geometry and load as coarse companion; 24x24 octant mesh.
Material / section	E=3e+06, nu=0.3; shell thickness=3. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	Rigid end diaphragm and octant symmetry; quarter of unit radial point load.
Applied load	Radial point force -0.25 in global X on the modeled octant symmetry corner.



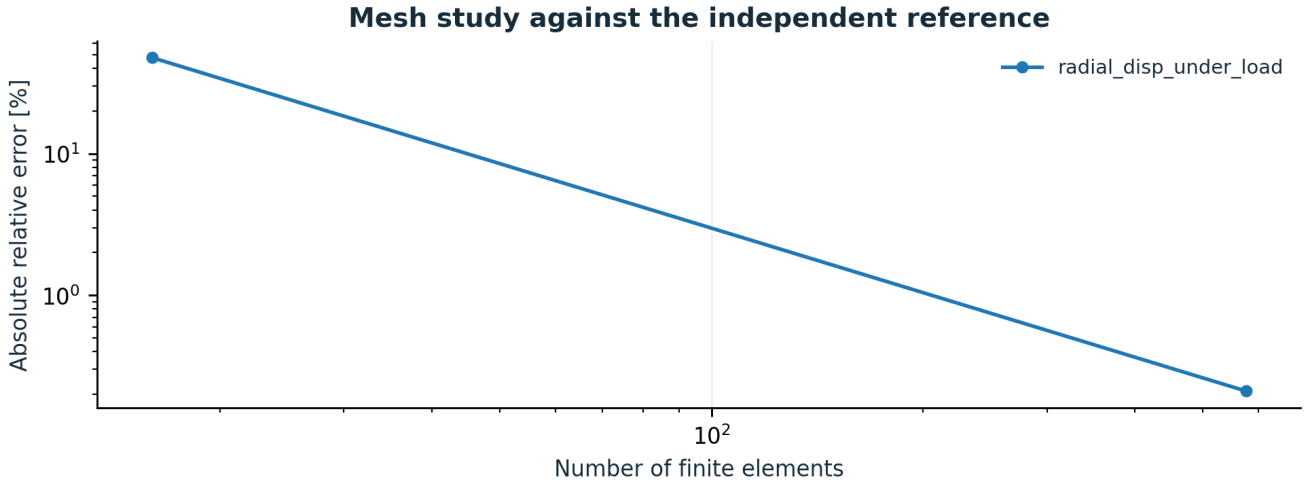
Shape amplified x1.97e+06; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
radial disp under load (magnitude) [benchmark length unit]	1.8286e-05	1.8248e-05	0.21088%	PASS

Evidence, convergence and interpretation



The separately retained MH_pinched_cylinder companion tests the same published target. Coarse-mesh error is displayed and is not relabeled as validation success.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MacNeal, R.H.; Harder, R.L. (1985). A proposed standard set of problems to test finite element accuracy. Finite Elements in Analysis and Design 1, 3-20. https://doi.org/10.1016/0168-874X(85)90003-4

Ko, Y.; Lee, Y.; Lee, P.-S.; Bathe, K.-J. (2017). Performance of the MITC3+ and MITC4+ shell elements in widely-used benchmark problems. Computers & Structures 193, 187-206. https://doi.org/10.1016/j.compstruc.2017.08.003

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv